Yusuf Miraç Uyar

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Education

Technical University of Munich — Munich, Germany
M.Sc. in Informatics — *Oct. 2023 – Present*
Current grade: 1.8 *(German grading scale; 1.0 = best)*

Bilkent University — Ankara, Turkey
B.S. in Computer Engineering, Full Merit Scholarship — *Sep. 2018 – 2023*
GPA: 3.64 / 4.00

Research Projects

Multi-Agent Reinforcement Learning for Flying Ad-Hoc Networks — Aug. 2026 – Present
Master's Thesis, Fraunhofer IKS — *Prof. Martin Bichler*

- Developed a **permutation-equivariant transformer policy** with sequential partner selection for decentralized optical aircraft networks in a use case introduced by a major aerospace company.
- Designed difference rewards for multi-agent credit assignment and compared **MAPPO/IPPO**, negotiation protocols, and synthetic versus real-flight training data.
- Built centralized optimization benchmarks using graph reachability and mixed-integer linear programming to quantify connectivity limits; evaluated on a full day of European flight data.
- On the **24-hour evaluation day**, the best learned policy achieved **90% of the nighttime connectivity upper bound**, compared with **73%** for an industry-method baseline, with comparable link stability.
- Currently investigating how aircraft density affects achievable connectivity and the traffic density required to maintain reliable paths to ground stations.

Digital Platform Detector — Winter 2025/26
Interdisciplinary Project, TUM — *Dr. David Angenendt*

- Fine-tuned **Mistral-7B with LoRA** on ORBIS/Crunchbase descriptions; achieved **0.82 F1** on 116 validation cases using literature-derived and expert-labelled training data.
- Constructed a **735-company dataset** by matching literature-derived platform companies to ORBIS, enriching descriptions with Crunchbase, and incorporating expert-labelled boundary cases.
- Refined classification instructions and negative examples, reducing the false-positive rate from **34.1% to 11.4%** and increasing precision from **74.1% to 91.4%** on the **116 validation cases**.

One Out Training for Physics Emulators Using Continuous Convolutions — Summer 2025
Guided Research, TUM — *Prof. Nils Thuerey*

- Evaluated deep learning for particle-based physical simulation across **three fluid datasets**, comparing continuous-convolution kernel bases, message-passing depth, and accuracy–compute trade-offs.
- Adapted LagrangeBench TGV-2D trajectories to the SFBC pipeline, computing missing SPH density fields, particle areas, and support radii while preserving periodic boundaries.
- Developed a **hybrid MSE–correlation loss** to improve the visual fidelity of predicted density fields. In a LagrangeBench TGV-2D comparison, reduced density-prediction MSE by **94.4%** and increased spatial correlation from **0.955 to 0.999**, without extra inference cost.
- One Out Training for Physics Emulators Using Continuous Convolutions (PDF)

Benchmarking Provably Safe Reinforcement Learning — Winter 2024/25
Master's Practicum, TUM — *Prof. Matthias Althoff*

- Implemented **zonotope-based action-set representations**; investigated how geometry and dimensionality affect **PPO** with continuous **action masking** and **action replacement** in **MuJoCo**.
- Compared L1- and L2-based action constraints for **continuous robotic control**, examining how their geometry affects the admissible actions and the computational cost of safe exploration.
- Worked with **quadrotor dynamics** and **state-dependent** safety constraints through the team's 2D and 3D simulation benchmarks, experiment on how action representations affect safe reinforcement learning.
- Analysed how increasing action-space dimensionality restricts feasible control inputs and raises computational costs, motivating more scalable L2-based zonotope representations.
- Benchmarking Provably Safe Reinforcement Learning (PDF)

Experience

Fraunhofer IKS Garching, Germany
Master's Thesis Research *Aug. 2026 – Present*
- Exploring classical reinforcement learning for flying ad hoc networks within a project comparing classical and quantum-based reinforcement learning approaches.

Olympos Istanbul, Turkey
Data Scientist – Part-Time *June 2024 – Jan. 2026*
- Built **LLM**-based models to analyze companies and predict VC firms' selection behavior with **73%** accuracy.
- Built early-stage **Y Combinator** detection models by combining **LLMs** with LinkedIn profile analysis.
- Enhanced a salary prediction pipeline by improving **data integration** across Glassdoor, Levels.fyi, and Crunchbase through a **Python**-based seniority inference model and **multithreaded processing**.

Data Scientist – Part-Time *June 2022 – May 2023*
- Developed a tokenization method in **Python** and **Pandas** that reduced Google Places API usage by **32%**.
- Identified and corrected **3%** of database record errors related to bias and network issues using **PostgreSQL**.
- Developed and deployed company quantification analyses using LinkedIn profile data.

Bilkent University Ankara, Turkey
Undergraduate Research Assistant *July 2021 – Mar. 2023*
- Researched StyleGAN architectures for kinship face generation in **PyTorch**, supervised by Dr. Hamdi Dibeklioğlu.

SBI Bilişim A.S. Ankara, Turkey
Software Engineering Intern *June 2020 – September 2020*
- Implemented functionalities to visualize trained convolutional neural networks using **Keras** and **TensorFlow**.
- Used those methods to combine trained networks that improved object detection by **8%** in the final product.

Bachelor Projects

Tutorium Fall 2022
TypeScript, React, HTML, CSS, SemanticUI
- Developed a platform that matched students with tutors who excelled in their subjects.
- Built the UI and backend integration using **TypeScript**, **React**, **CSS**, and **SemanticUI**.
- Information Page

BilHub Fall 2021
JavaScript, React, HTML, CSS, SemanticUI
- Developed a group project management application to organize teams, manage tasks, and review artifacts.
- Designed and developed the application's UI using **CSS** and **SemanticUI**.
- Developed application functionality and database connection using **JavaScript** and **React**.
- GitHub Page

Skills

Programming: Python, C/C++, SQL
ML: PyTorch, MARLlib, Gymnasium, Tianshou, scikit-learn
Tools: Git, PostgreSQL
Languages: English (Proficient), Turkish (Native), German (Beginner)

Academic Achievements

Turkish National Olympiad of Informatics – Bronze Medal 2017